# Duane Pinkerton

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Please explore my interactive resume at hire-duane.org.

## Summary

I've spent my career in technical professional services, doing hands-on implementation work directly with customers. My approach is to sit with a customer, learn what they need, find where the product falls short, and write the missing piece. I've built data pipelines and integration systems from scratch and owned them in production at enterprise scale, including the architecture decisions as they grew. More recently I've brought the same approach to AI, building agentic systems on LangGraph, MCP, and RAG, with model output validated in code. I'm drawn to understanding how systems work, and to making them work better.

## Skills

**Customer engagement:** Requirements discovery · Gap analysis · Translating technical constraints into business terms · Knowing when to build vs. configure

**Languages & Scripting:** Python · Java · SQL · JavaScript · Unix/Shell · REST APIs

**Infrastructure:** Docker · Kubernetes · AWS (EC2) · Oracle Cloud Infrastructure (OCI) · Linode · Git · Jenkins · CI/CD

**Technical strengths:** SQL performance tuning · Data pipeline architecture · Distributed systems operations · System integration

**LLM integration:** Claude (Anthropic), OpenAI, multi-provider routing; RAG patterns, vector search, pgvector, semantic embeddings, Supabase

**Agentic systems:** LangGraph (orchestration, checkpointing, human-in-the-loop), MCP (Model Context Protocol, consuming + publishing servers); Jev (TypeSafe AI decision model)

**Prompt engineering:** System prompt design, context engineering, code-based validation of LLM output, evaluation, LLM cost optimization

**Tooling & production:** Langfuse (observability, traces, prompt versioning), n8n (workflow automation), deployment, security threat modeling

**Edge & on-device AI:** wake-word detection, speech-to-text, and text-to-speech on Raspberry Pi (openWakeWord, Moonshine, Piper)

## Production AI Projects (2024–present)

- **Built and deployed an agentic email pipeline:** a LangGraph agent that sorts job emails with a decision model, extracts posting links with an LLM whose output is validated in code, and hands ambiguous cases to a human over Slack. Moved the sorting decision from an LLM to Jev, which cut cost per email from about $1.00 to $0.04 and processing time from minutes to seconds. Langfuse for observability, MCP servers on both ends. Wrote the threat model before writing any code, with a specific mitigation for prompt injection, stored XSS, and cross-tenant access
- **Designed and built private voice assistant devices** on Raspberry Pi and ReSpeaker hardware: wake word, speech-to-text, and speech synthesis all run on the device itself, feeding a self-hosted AI agent hands-free. Each unit keeps its own memory scope and listens on no inbound ports. Brought the same design to the desktop as Agent Voice, an open-source voice interface for Omarchy: say a wake word and whichever AI agent is installed (Claude Code, Codex, Gemini) answers out loud, with all speech processing on the machine, tuned to run CPU-only on a 2014 laptop
- **Launched two production AI sites.** hire-duane.org is an interactive resume with RAG-backed chat and job-fit evaluation, built to be readable by crawling agents. job-radar.net runs on AWS and handles multi-provider job search with AI scoring, resume tailoring, application workflow, and interview prep
- **Deployed OpenBrain behind custom MCP servers:** a model-agnostic memory layer on PostgreSQL and pgvector that answers semantic queries from Claude, ChatGPT, Cursor, or any MCP client, all against one shared knowledge base
- **Automated multi-step workflows in n8n,** wiring APIs and services together with LLM-powered decision nodes

## Experience

### Responsys / Oracle

2007 – 2026

*Principal Professional Solutions Consultant → Data Architect → Solutions Architect → Principal Applications Engineer, Operations*

**Owning the Platform (2013 – 2026)**

- Fielded the final escalations on production data integration issues as the framework's author, joining customer calls as the subject matter expert and advising on how their integrations should be built
- Made the architectural decisions on reliability, redundancy, and capacity planning after Oracle acquired the company
- Deployed the data integration environment in new datacenters as the platform expanded
- Led the migration of the production data integration environment from bare metal to containers on Oracle Cloud Infrastructure (OCI), using Docker, Kubernetes, Git, and Jenkins, and finished it without a significant downtime event

**Building the Platform (2007 – 2013)**

- Designed and built the company's data pipeline and integration framework from scratch. It replaced 95% of the custom scripts running in production, became the standard across professional services, and grew to carry 200+ production jobs for 80+ enterprise customers
- Trusted with the company's highest-profile customer engagements, owning delivery end to end: discovery, requirements gathering, documentation, implementation, custom development, and handover to support
- Trained consultants and partners on implementation best practices; primary subject matter expert for custom data integration across the organization

**Notable customers:** Williams Sonoma, Pottery Barn, West Elm, Epson, Orbitz, Carlson Wagonlit, Verizon, Disney, Symantec, Lenovo, Levi's, Lego, PayPal, See's, Coach, Grainger, Intuit, Wells Fargo, BabyCenter

### Extensity / Geac / Infor

1999 – 2007

*Principal Professional Services Consultant → Team Lead, Custom Solutions Group*

- Ran enterprise software implementations start to finish, from discovery and requirements through training and support handover, for domestic and international clients
- Built multi-currency and exchange rate support from scratch, which made the company's first international enterprise deployments possible at Cisco and Sybase. The features later shipped in the core product
- Worked directly with engineering to drive bug fixes and code changes into the product codebase
- Spent 18 months in the UK as lead technical member of the founding professional services team; trained and mentored three colleagues, hands-on across all client implementations
- Turned around six at-risk implementations led by other consultants, diagnosing deployment and implementation issues and getting each customer back on track
- Led a team of professional services developers implementing custom solutions across the client base

**Notable customers:** Cisco Systems, Sybase, Ernst & Young, A.T. Kearney, State of Alaska, Franklin Investments, Lawrence Livermore National Labs, Memorial Sloan Kettering Cancer Center, UCLA, McKinsey & Co.

## Education

**Bachelor of Arts in Computer Science** · UC Berkeley